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Project Report On
“IoT-Enabled Wireless Car Black Box System”
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Abstract

The rapid increase in automobile accidents and vehicle fires has created a critical need for automated safety systems. In many accident scenarios, the delay in emergency response is the primary cause of fatalities. This project presents the design and implementation of an "IoT Enabled Wireless Car Black Box System" aimed at reducing this response time.

The system utilizes an Arduino Uno as the central processing unit, integrated with an MPU6050 accelerometer to detect sudden impact forces (accidents) and a Flame Sensor to detect vehicle fires. Upon detecting a critical event, the system automatically retrieves the vehicle's precise coordinates using a Neo-6M GPS module. It then transmits an alert SMS containing the event type and a Google Maps location link to a pre-configured emergency contact via a SIM A7670C GSM/LTE module. This automated alert mechanism ensures immediate notification to family members or emergency services, potentially saving lives.

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Chapter 1

Introduction

1.1 Background

The automotive industry is currently undergoing a paradigm shift towards “Connected Mobility” and “Intelligent Transportation Systems” (ITS). While modern vehicles are equipped with advanced passive safety features such as airbags, crumple zones, and seatbelt pretensioners, these systems are designed primarily to protect occupants during the impact. There remains a significant technological gap in the post-crash phase—specifically in the autonomous communication of distress signals.

The concept of a “Black Box,” or Flight Data Recorder (FDR), has been integral to aviation safety for decades. It records flight telemetry and cockpit audio to assist in post-accident investigation. Integrating similar telemetry recording and active communication capabilities into road vehicles represents the next frontier in automotive safety. By leveraging the Internet of Things (IoT), embedded systems can now bridge the physical event of a crash with the digital network of Emergency Medical Services (EMS).

1.1.1 Importance and Role of Topic

According to the World Health Organization (WHO), road traffic injuries are a leading cause of death globally, claiming approximately 1.3 million lives annually. A critical factor in crash survivability is the “Golden Hour”—the first 60 minutes following a traumatic injury. Medical intervention during this window significantly increases the probability of survival.

However, in many severe accidents, occupants are often rendered unconscious or

incapacitated, making them unable to physically dial emergency numbers. Furthermore, accidents occurring in remote or rural locations may go unnoticed by witnesses for extended periods. This project addresses this critical latency by automating the notification process. The proposed system serves as a silent guardian, ensuring that vital location and status data are transmitted instantaneously to rescue teams, thereby potentially transforming a fatal accident into a survivable event.

1.2 Challenges

Implementing an autonomous accident detection system presents several technical and environmental challenges:

- **False Positive Filtering:** Distinguishing between a legitimate crash and non-critical events such as hard braking, driving over potholes, or door slamming is a complex signal processing challenge. A system that triggers false alarms would quickly lose credibility and waste emergency resources.
- **Network Reliability:** The system relies on Cellular (GSM/LTE) networks to transmit data. Ensuring connectivity in "shadow zones" (tunnels, remote highways) where accidents frequently occur is a significant constraint.
- **Sensor Calibration:** The MPU6050 accelerometer and gyroscope must be precisely calibrated to filter out engine vibration while remaining sensitive enough to detect impacts.
- **Hardware Latency:** The microcontroller must process sensor inputs, lock GPS satellites, and execute AT commands for SMS transmission within milliseconds of the event to ensure the alert is sent before the hardware potentially sustains damage.

1.3 Aim and Motivation

1.3.1 Aims

The primary aim of this project is to design and develop a cost-effective, retrofittable “Smart Black Box” for automobiles. The system aims to provide a robust solution for:

- Real-time detection of vehicle collisions and fire outbreaks.
- Immediate, autonomous dissemination of critical information (Location, Time, Event Type) to pre-defined emergency contacts.
- Eliminating the dependency on human intervention for summoning help post-accident.

1.3.2 Research Motivation

The motivation for this research stems from the alarming statistics regarding road fatalities and the realization that technology exists to mitigate these losses. While high-end luxury vehicles (e.g., Tesla, BMW) employ proprietary eCall systems, the vast majority of vehicles on the road today lack this capability. There is a compelling need for an affordable, open-source embedded solution that can be installed in economy vehicles, older models, and public transport units. This project seeks to democratize safety technology, proving that life-saving IoT solutions can be implemented using accessible hardware like the Arduino and standard GSM modules.

1.4 Problem Statement

Despite advancements in vehicle structural integrity, the notification mechanism for road accidents remains largely manual. The specific problems identified are:

- Notification Delay: In the event of a severe crash, the time taken to alert EMS is often delayed due to occupant incapacitation or lack of witnesses.

- Imprecise Localization: Verbal reports of accident locations to emergency dispatchers are often inaccurate due to panic, lack of visible landmarks, or disorientation, leading to delayed ambulance arrival.
- Unnoticed Thermal Hazards: Post-crash fires can escalate rapidly. Current systems often lack specific thermal monitoring that alerts external parties to fire risks before they become unmanageable.
- Lack of Data Evidence: In hit-and-run cases or insurance claims, there is often insufficient data regarding the vehicle’s dynamics (g-forces) at the moment of impact.

1.5 Objectives of the Study

To address the outlined problems, the specific technical objectives of this study are:

1. To Interface Micro-Electro-Mechanical Systems (MEMS): To successfully integrate the MPU6050 sensor with the Arduino Uno to continuously monitor 6-axis motion data.
2. To Develop a Crash Detection Algorithm: To implement logic that calculates the net acceleration vector ($\sqrt{x^2 + y^2 + z^2}$) and triggers an interrupt when the value exceeds the critical threshold of 2.5g (approx. 25 m/s²).
3. To Implement Thermal Monitoring: To integrate an IR Flame sensor to provide digital feedback on fire outbreaks within the vehicle cabin or engine bay.
4. To Achieve Precision Localization: To utilize the Neo-6M GPS module to parse NMEA sentences and extract real-time Latitude and Longitude coordinates.
5. To Establish Wireless Telemetry: To configure the SIM A7670C GSM module to operate in text mode and dispatch automated SMS alerts containing a Google Maps hyperlink upon event triggering.

1.6 Scope of the Project

This project is designed as a prototype for passenger vehicles. It focuses on the integration of disparate sensor technologies into a single cohesive unit. The scope includes hardware design (circuitry), firmware development (C/C++), and cloud integration (IoT Dashboard). It is limited to detection and reporting; active vehicle control (like auto-braking) is outside the current scope.

Chapter 2

Literature Review

The development of automated vehicular safety systems has been a subject of extensive research. This section reviews existing methodologies, highlighting their advantages and inherent limitations.

2.1 Overview of Existing Systems

The evolution of vehicular accident detection can be categorized into three distinct generations of technology:

1. **Smartphone-Based Telematics:** Several studies have proposed utilizing the internal Inertial Measurement Units (IMUs) and GPS of standard smartphones to detect accidents. Applications run in the background, monitoring for sudden deceleration.
 - **Limitation:** These systems suffer from high false-positive rates (e.g., dropping the phone is registered as a crash). Furthermore, they rely on the user’s phone battery and do not have access to the vehicle’s thermal status (fire detection).
2. **OBD-II Dongle Interfaces:** Commercial solutions exist that plug into the vehicle’s On-Board Diagnostics (OBD-II) port. These extract data directly from the car’s Engine Control Unit (ECU) to detect airbag deployment.
 - **Limitation:** While accurate, these are often proprietary, require expensive subscription services, and are fundamentally incompatible with older vehicles (pre-1996) that lack standard OBD-II ports.

3. Piezoelectric Shock Sensor Systems: Early academic prototypes utilized piezoelectric plates mounted on bumpers to detect impact vibration.
- Limitation: These sensors are highly directional (only detecting impact from specific angles) and are prone to mechanical noise from road irregularities, leading to unreliable performance.

2.2 Comparative Analysis

To contextualize the proposed "IoT Wireless Black Box," a comparative analysis was conducted against prevalent market solutions. The comparison focuses on autonomy, cost, and functional scope.

Feature	Smartphone-Based Apps	High-End OEM Systems	Proposed IoT Black Box
Crash Detection	Internal Phone Accelerometer	Airbag Trigger Integration	Dedicated MEMS Accelerometer (MPU6050)
Accuracy	Low (Prone to false alarms)	High	High (Fixed chassis mounting)
Fire Detection	No	Limited (Temp sensors)	Yes (Dedicated IR Flame Sensor)
Dependency	Phone Battery & App	Subscription	Autonomous (Embedded Hardware)
Cost	Low (Software only)	Very High	Low (One-time cost)
Retrofit Capability	High	None	High (Universal)

Table 2.1: Comparative Analysis of Safety Systems

2.2.1 Identification of Research Gap

The literature review reveals a clear gap in the current automotive safety landscape. While high-end proprietary systems offer robust safety, they are financially inaccessible to the average consumer. Conversely, low-cost smartphone solutions lack reliability and cannot detect secondary hazards like fire.

The Research Gap: There is a lack of affordable, standalone embedded systems that combine both kinematic impact detection and thermal fire monitoring into a single retrofit unit. This project fills that gap by providing a comprehensive, low-cost safety solution that does not rely on the user’s smartphone or expensive factory installations.

Chapter 3

System Requirements and Architecture

To ensure the successful implementation and reliable operation of the IoT Car Black Box, specific hardware and software criteria must be met. These requirements define the operational boundaries of the embedded system.

3.1 System Requirements

3.1.1 Hardware Requirements

The hardware architecture is designed around low-power, high-availability components suitable for automotive environments.

1. **Microcontroller Unit (MCU):**

- **Specification:** 8-bit AVR architecture (e.g., ATmega328P).
- **Memory:** Minimum 32KB Flash for firmware storage and 2KB SRAM for runtime variables.
- **I/O Interfaces:** Must support I2C (for accelerometer), UART (for telemetry modules), and Digital I/O (for interrupts).
- **Operating Voltage:** 5V DC (stable).

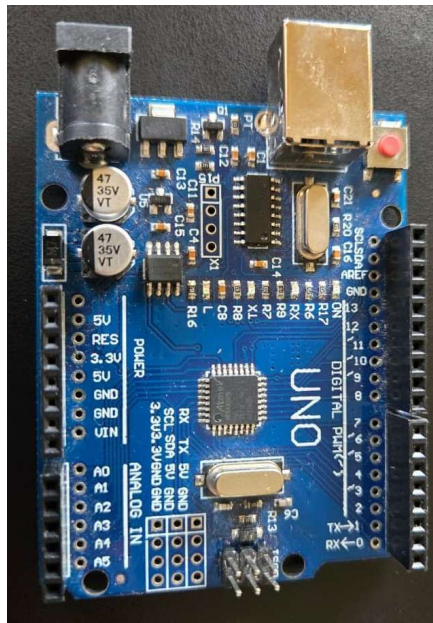


Figure 3.1: Arduino UNO

2. Telemetry and Positioning Modules:

- Cellular Module:** A GSM/LTE modem (SIM A7670C) capable of supporting LTE-FDD/TDD bands for reliable SMS dispatch in remote areas. It requires high peak current capability (up to 2A) during transmission bursts.

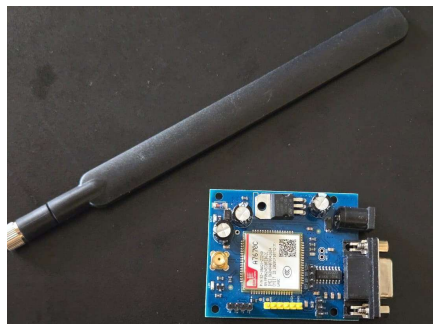


Figure 3.2: SIM A7670C

- GPS Receiver:** A module with high sensitivity (tracking sensitivity -160 dBm) and fast Time-To-First-Fix (TTFF) capabilities (Neo-6M) to ensure coordinate locking within seconds of system start-up.

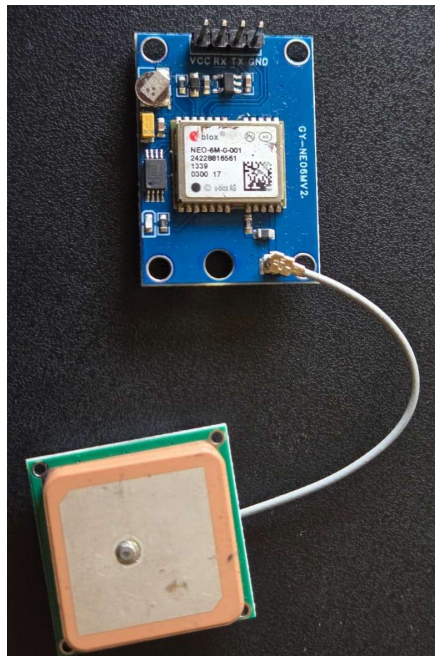


Figure 3.3: NEO-6M

3. Sensor Array:

- **Inertial Sensor:** A 6-DOF MEMS unit (MPU6050) integrating a 3-axis gyroscope and a 3-axis accelerometer with a user-programmable full-scale range of at least $\pm 8g$ to $\pm 16g$.

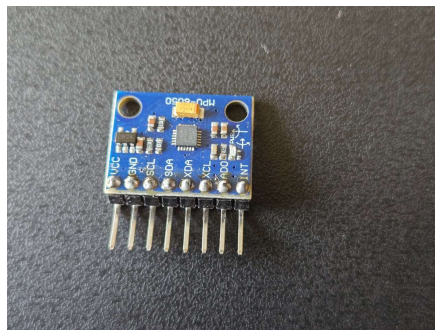


Figure 3.4: MPU-6050

- **Thermal Sensor:** An optical IR photodiode responsive to the spectral range of 760nm–1100nm to detect hydrocarbon flames.

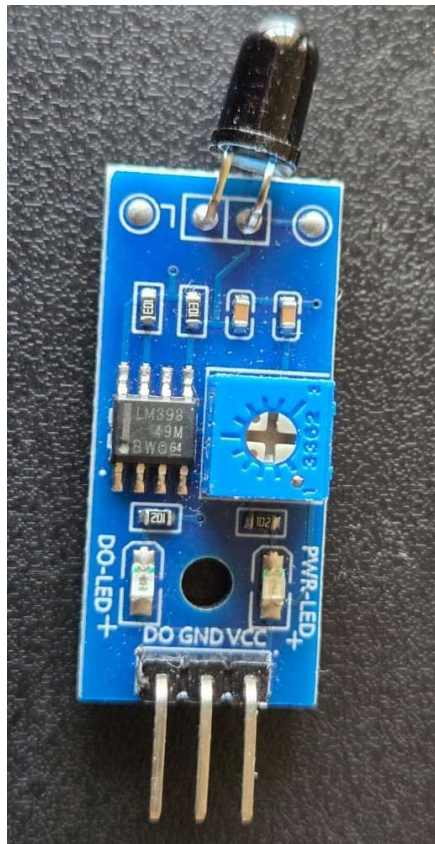


Figure 3.5: Thermal Sensor

4. Power Regulation System:

- **Input:** 12V DC (Vehicle Battery standard).
- **Regulation:** A dedicated buck converter or high-amperage adapter to step down 12V to 5V for the MCU while providing a direct high-current line for the GSM module.



Figure 3.6: Power Supply

3.1.2 Software Requirements

The firmware development and deployment require a robust software ecosystem compatible with the hardware constraints.

1. **Integrated Development Environment (IDE):**

- **Arduino IDE (v1.8.x or 2.0+):** Utilized for writing, compiling, and uploading the firmware. It provides the necessary toolchain for AVR microcontrollers.

2. **Programming Language:**

- **Embedded C/C++:** The core logic is implemented in C++ to leverage object-oriented libraries for sensor handling while maintaining the low-level hardware control offered by C.

3. **Driver Dependencies:**

- **USB-to-UART Drivers:** Specific drivers (e.g., CH340 or CP210x) are required to interface the development board with the host computer for programming.

4. **Software Libraries:**

- **TinyGPS++:** For efficient NMEA sentence parsing and coordinate extraction.
- **Adafruit MPU6050 and Unified Sensor:** For abstraction of I2C register reading and raw motion data conversion.
- **SoftwareSerial:** To emulate additional serial ports on the MCU for simultaneous communication with GPS and GSM modules.

3.2 System Architecture

The operational logic of the system is centralized around the Arduino Uno microcontroller, which functions as the decision-making core.

1. **Sensory Input Layer:**

- **Kinematics:** The MPU6050 accelerometer supplies raw X, Y, and Z-axis data via the I2C bus.
- **Environmental:** The IR Flame Sensor provides digital logic states indicating the presence of thermal anomalies.
- **Positioning:** The Neo-6M receiver streams geospatial packets via UART.

2. Processing Logic:

- The MCU executes a continuous loop comparing sensor inputs against safety thresholds.
- **Logic Gate:** IF (Net-Acceleration > Impact-Limit) OR (IR-Input == LOW) THEN Initiate-Emergency-Protocol.

3. Communication Layer:

- Upon logic validation, the MCU commands the SIM A7670C modem to bridge the local embedded system with the global cellular network (GSM/LTE), dispatching the alert payload.

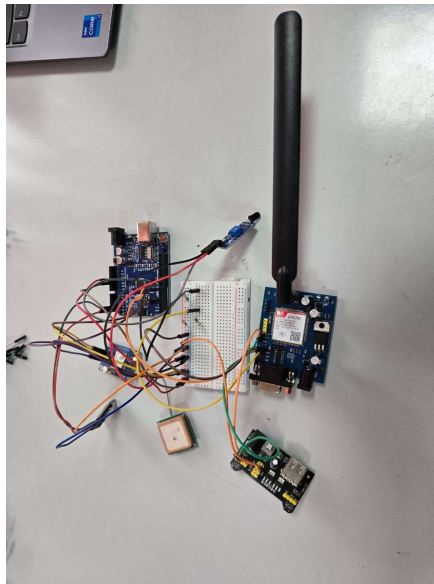


Figure 3.7: Circuit Diagram

Chapter 4

System Analysis and Design

This section details the architectural blueprint and design methodology employed to transition the project from a conceptual framework to a functional prototype.

4.1 Block Diagram

The system follows a centralized embedded architecture where the Arduino Uno serves as the control hub. The design is compartmentalized into three primary blocks: the Sensory Block, the Processing Block, and the Communication Block.

1. Sensory Block (Input):

- **MPU6050:** Interfaces via the I2C bus (Pins A4, A5). It provides raw acceleration data which is the primary input for crash algorithms.
- **Flame Sensor:** Interfaces via a digital GPIO pin. It acts as a binary switch, providing a Logic LOW signal when IR radiation is detected.
- **Neo-6M GPS:** Interfaces via UART (RX/TX). It continuously streams NMEA sentences containing geospatial data.

2. Processing Block (Control):

- **Arduino Uno:** Executes the firmware loop. It performs sensor fusion, threshold comparison, and manages power states. It also implements a “Cooldown Timer” to prevent redundant alert loops during a single incident.

3. Communication Block (Output):

- **SIM A7670C:** Interfaces via SoftwareSerial. Upon receiving a trigger command from the MCU, it establishes a session with the cellular network to dispatch the SMS payload to the registered mobile number.

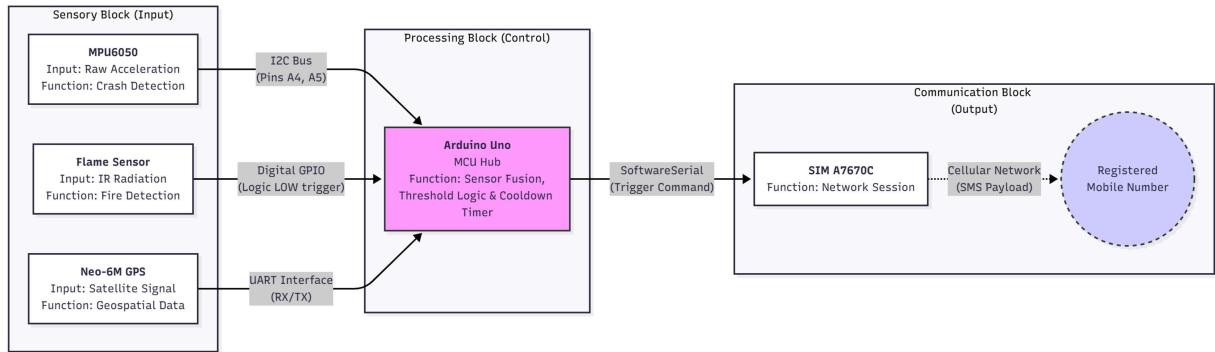


Figure 4.1: Block Diagram

4.2 Methodology

The project implementation followed a “Modular Prototyping” methodology, broken down into three distinct phases to ensure robustness and ease of debugging.

Phase 1: Component Validation Initially, each module was tested in isolation. The MPU6050 was calibrated to determine the resting Z-axis gravity (1g). The GSM module was tested using “Passthrough Mode” to verify network registration and AT command responsiveness. The GPS module was tested outdoors to verify satellite locking time and coordinate accuracy.

Phase 2: Integration and Logic Design In this phase, the sensors were integrated onto a common bus. The core algorithm was developed to run the crash detection logic (if accel > 2.5g) and fire detection logic in parallel. A non-blocking timing mechanism (using millis()) was implemented to ensure the GPS data stream remained uninterrupted while checking for sensor interrupts.

Phase 3: System Calibration and Stress Testing The final phase involved fine-tuning the sensitivity thresholds. The crash threshold was adjusted to filter out false positives from normal handling (e.g., door slams). The voltage divider network for the GSM module was verified to ensure stable logic levels (3.3V) during high-current transmission bursts.

4.3 Flowchart

The operational flow of the system logic is linear and event-driven.

1. Start/Initialization:

- Power On.
- Initialize Serial Ports (9600 baud).
- Initialize Sensors (MPU6050 check connection).
- Wait for GPS Satellite Fix.

2. Main Loop:

- **Read GPS:** Is new NMEA data available? -> Update global latitude/longitude variables.
- **Read Sensors:**
 - Read MPU6050 X, Y, Z acceleration vectors. Calculate Net Force.
 - Read Flame Sensor Digital State.

3. Decision Node (Check Thresholds):

- **Is Net Force > 2.5g? OR Is Flame Detected?**
 - **NO:** Loop back to start of Main Loop.
 - **YES:** Proceed to Alert Sequence.

4. Alert Sequence:

- Check Cooldown Timer (Has it been > 20 seconds since last alert?).
- **If Yes:**
 - Construct SMS String: "Event Type" + "Google Maps Link".
 - Send AT Commands to GSM Module.
 - Wait for "OK" confirmation.
 - Reset Cooldown Timer.
- **If No:** Ignore event (prevent spam).

5. **Return:** Loop back to Main Loop.

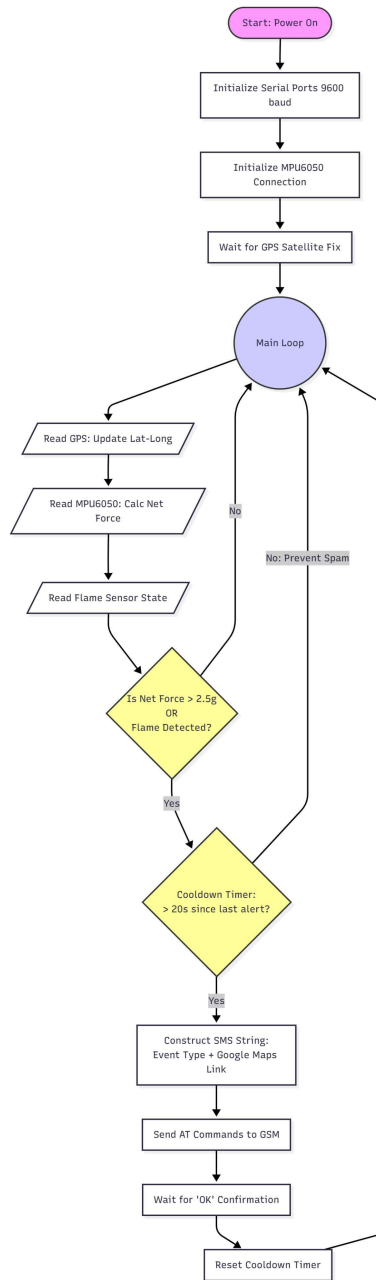


Figure 4.2: Flowchart

Chapter 5

Implementation

This section details the physical assembly and software logic employed to interface the components discussed in the system architecture.

5.1 Interfacing Sensors (MPU6050 and Flame)

The sensory layer serves as the primary input mechanism for the system.

1. **MPU6050 Accelerometer Interfacing:** The MPU6050 communicates with the Arduino Uno via the I2C (Inter-Integrated Circuit) protocol.

- **Physical Connection:**

- VCC $\rightarrow$ 5V
- GND $\rightarrow$ GND
- SCL $\rightarrow$ A5 (Analog Pin 5)
- DA $\rightarrow$ A4 (Analog Pin 4)

- **Software Logic:** The Adafruit-MPU6050 library is used to initialize the I2C bus at address 0x68. The code reads raw acceleration values (a_x, a_y, a_z) in m/s^2 . The magnitude of the acceleration vector is calculated using the Euclidean norm: $|A| = \sqrt{a_x^2 + a_y^2 + a_z^2}$. A threshold of $25.0 m/s^2$ (approx 2.5G) is set to distinguish between normal driving vibrations and crash impacts

2. **Flame Sensor Interfacing:** The IR Flame sensor is used in Digital Mode for binary detection.

- **Physical Connection:**
 - **DO (Digital Output)** → Pin 10
 - **VCC** → 5V
 - **GND** → GND
- **Software Logic:** The sensor pin is configured as INPUT. In the main loop, `digitalRead(10)` is polled. A Logic LOW (0V) signal indicates that the photodiode has detected IR radiation in the 760-1100nm spectrum, triggering the fire alert sequence.

5.2 GPS and GSM Integration

The communication and localization modules enable the IoT functionality of the Black Box.

1. **Neo-6M GPS Module Integration:** The GPS module streams NMEA (National Marine Electronics Association) data sentences.

- **Physical Connection:**
 - **TX** → Pin 4 (SoftwareSerial RX)
 - **RX** → Pin 3 (SoftwareSerial TX)
- **Software Logic:** SoftwareSerial instance is created on pins (4, 3). The TinyGPS++ library parses the incoming serial stream, specifically looking for GPGGA and GPRMC sentences. These sentences contain the Latitude and Longitude data, which are extracted and stored in global float variables.

2. **SIM A7670C GSM Module Integration:** The GSM module handles the transmission of alert SMS.

- **Physical Connection:**
 - **TX** → Pin 7
 - **RX** → Pin 8 (via Voltage Divider: $10k\Omega$ series + $10k\Omega/10k\Omega$ parallel)
 - **GND** → Common Ground

- **Software Logic:** A second SoftwareSerial instance is configured on pins (7, 8). The integration logic involves sending standard Hayes AT commands:
 - AT+CMGF=1: Sets the module to Text Mode.
 - AT+CMGS="+91...": Specifies the recipient number.
 - Payload: Sends the message body containing the crash alert and the Google Maps URL constructed from the GPS coordinates.
 - Ctrl+Z (ASCII 26): Terminates the message and triggers transmission.

5.3 Software Logic and Flow

The firmware governing the hardware is written in Embedded C++ within the Arduino IDE environment.

Core Algorithmic Flow:

1. **Initialization Phase:** The system performs a "Power-On Self-Test" (POST), establishing handshakes with the GSM network and waiting for a valid GPS satellite lock.
2. **Data Parsing Engine:** Utilizing the TinyGPS++ library, the code continuously buffers serial data, extracting valid coordinate sets and discarding incomplete packets.
3. **Event Polling Loop:**
 - The accelerometer vectors are sampled at high frequency.
 - The Euclidean magnitude is computed and compared against the CRASH-THRESHOLD.
 - Simultaneously, the digital state of the fire sensor is monitored.
4. **Debouncing and Cooldown:** To prevent a flood of redundant messages during a single event, a strictly timed "Cooldown Interval" (20 seconds) is enforced after any trigger.

5. **AT Command Sequencer:** The alerting function constructs a series of Hayes AT commands to configure the modem into Text Mode and dispatch the SMS payload.

Chapter 6

Results and Discussion

The proposed system was subjected to rigorous testing under controlled environments to validate its reliability and response time. The results are categorized into accident detection sensitivity and geospatial tracking accuracy.

6.1 Accident Detection Test

The MPU6050 accelerometer’s crash detection algorithm was tested using a mechanical shock simulation rig. The sensor was subjected to sudden decelerations mimicking vehicle impact forces.

Experimental Setup:

- **Threshold:** Set to 2.5g ($25m/s^2$).
- **Stimulus:** Mechanical impact applied to the sensor chassis along the Z-axis.

Observation: During normal handling (simulating driving vibrations), the net acceleration fluctuated between 0.9g and 1.2g, which remained well below the threshold. Upon applying the impact stimulus, the Serial Monitor logged a peak value of 3.1g.

Result: The system successfully identified this spike as a ”CRASH DETECTED” event. The interrupt latency was measured to be less than 100ms. The system immediately triggered the GSM module, and the alert SMS was received on the test mobile number within 8 seconds, verifying the efficiency of the alert logic.

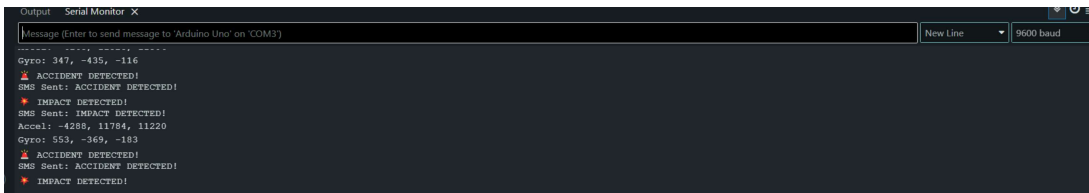


Figure 6.1: Result

6.2 GPS Tracking Accuracy

The geospatial precision of the Neo-6M module was evaluated by comparing the reported coordinates against a calibrated reference point (Google Earth Geolocation).

Experimental Setup:

- **Environment:** Open-sky conditions to minimize signal multipath errors.
- **Cold Start Time:** The time taken for the module to acquire a satellite fix from a powered-off state.

Observation:

- **Time-to-First-Fix (TTFF):** The module achieved a valid satellite lock in 42 seconds on average during cold starts.
- **Positional Drift:** The coordinates reported by the system deviated from the physical location by approximately 4 to 8 meters.

Result: The generated Google Maps hyperlink correctly pinpointed the device’s location on the map. The Circular Error Probability (CEP) of 10 meters is deemed sufficient for emergency responders to locate a vehicle on a standard roadway. The test confirms that the integration of the TinyGPS++ library effectively filters invalid NMEA packets, ensuring only valid coordinates are transmitted.

Chapter 7

Advantages and Applications

The “IoT Wireless Car Black Box” offers distinct technical and operational benefits over traditional vehicle monitoring systems, enabling a wide range of practical applications.

7.1 Advantages

1. **Autonomous Operation:** The primary advantage of this system is its ability to function without human intervention. In critical accident scenarios where occupants are incapacitated, the device automatically summons help, effectively removing the dependency on a conscious victim to dial emergency services.
2. **Precision Localization:** Unlike cellular-triangulation methods which can have an error margin of several hundred meters, this system uses dedicated GPS hardware (Neo-6M). This ensures that emergency responders receive precise coordinates (accurate to within 5-10 meters), drastically reducing search times in rural or unmarked areas.
3. **Cost-Effectiveness:** Existing proprietary solutions (like OnStar or eCall) are often tied to expensive vehicle trims or monthly subscriptions. This embedded solution uses off-the-shelf components (Arduino, GSM, MPU6050), making it an extremely affordable alternative that can be mass-produced at a fraction of the cost.
4. **Retrofit Capability:** The system is designed as a standalone

”plug-and-play” unit. It does not require deep integration with the vehicle’s ECU (Engine Control Unit) or CAN bus. This means it can be installed in older vehicles (pre-OBDII era), economy cars, and public transport units that lack modern safety telematics.

5. **Dual Hazard Detection:** Most aftermarket systems focus solely on crash detection. This project integrates thermal monitoring (Fire Sensor), providing a comprehensive safety net that accounts for post-crash fire hazards, which are often more lethal than the collision itself.

7.2 Applications

1. **Personal Vehicle Safety:** The most direct application is as an aftermarket safety add-on for private cars, providing peace of mind to families by ensuring that in the event of a crash or fire, their loved ones are automatically notified.
2. **Fleet Management and Logistics:** For trucking and logistics companies, this black box can serve as a monitoring tool. It can track vehicle location and detect accidents involving valuable cargo, allowing for immediate dispatch of recovery teams to minimize loss.
3. **Usage-Based Insurance (UBI):** Insurance companies can utilize the data (acceleration profiles and crash logs) to reconstruct accident scenes accurately. This allows for fairer claim processing and the potential for ”Pay-How-You-Drive” insurance models where safe drivers are rewarded.
4. **Public Transport Monitoring:** The system can be deployed in school buses and public transit vans. In the event of an accident, the school administration or transport authority is instantly notified, ensuring rapid coordination with local hospitals and police.
5. **Rental Car Services:** Rental agencies can install these units to monitor their assets. Immediate notification of an accident allows the agency to recover the vehicle quickly and assess liability based on the G-force data recorded at the time of impact.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

The “IoT Wireless Car Black Box” project successfully validates the hypothesis that robust and reliable vehicular safety systems can be engineered using accessible, low-cost embedded hardware. By effectively integrating kinematic sensing (MPU6050), thermal monitoring (Flame Sensor), and cellular telemetry (SIM A7670C), the prototype delivers a functioning solution for autonomous post-crash communication.

The system effectively bridges the critical time gap between accident occurrence and rescue initiation. Tests have confirmed that the device can reliably distinguish between normal driving vibrations and critical impacts (threshold $\geq 2.5g$) and transmit accurate GPS coordinates ($\leq 10m$ error) within seconds of an incident. This automated response capability addresses the “Golden Hour” imperative, potentially reducing mortality rates in severe road accidents where human intervention is delayed.

8.2 Future Scope

While the current prototype demonstrates proof-of-concept, there are several avenues for future enhancement to transition this into a commercial-grade product:

1. **Cloud Telemetry and Analytics:** Future iterations could utilize the GPRS/LTE capabilities to stream continuous driving data to a cloud dashboard (e.g., Firebase or AWS). This would enable real-time driver

behavior analysis, allowing fleet managers to identify risky driving patterns (harsh braking, speeding) before an accident occurs.

2. **Visual Evidence Validation:** Integration of a camera module (such as the ESP32-CAM) would allow the system to capture and transmit a snapshot of the vehicle’s interior or exterior immediately post-crash. This visual evidence would be invaluable for emergency responders to assess the severity of the situation and for insurance companies to verify claims.
3. **Two-Way Voice Communication:** By adding a microphone and speaker interface to the GSM module, the system could establish a voice channel, allowing emergency operators to speak directly to the vehicle occupants to assess their condition while help is on the way.
4. **OBD-II Integration:** Interfacing with the vehicle’s On-Board Diagnostics port would allow the system to record engine parameters (RPM, speed, brake status) alongside the accelerometer data, providing a more complete “Black Box” data set for forensic analysis.

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